

Technical Indicators & Strategy Backtesting

Risk-adjusted comparison of algorithmic trading strategies on AAPL

Universe: AAPL | Lookback: 5 years daily OHLCV | Data source: Synthetic (fallback) | Generated: 2026-06-29

Initial capital \$100,000. Long-only, no leverage, no short selling. Signals computed on the daily close and executed at the next open; a 1 bp cost models slippage (Alpaca equity commissions are \$0).

1. Strategy descriptions & rules

Trend Following [Trend (MACD + ADX + SMA-50)]

Entry: MACD > signal AND ADX > 25 AND close > SMA-50

Exit: MACD < signal

Mean Reversion [Momentum + Volatility (RSI + Bollinger Bands)]

Entry: RSI < 30 AND close < lower Bollinger Band

Exit: RSI > 70 OR close > upper Bollinger Band

Custom Triple Confirmation [Trend + Momentum + Volume (EMA + RSI + CMF)]

Entry: EMA-50 > EMA-200 AND RSI crosses up through 50 (<70) AND CMF > 0

Exit: close < EMA-50 OR RSI > 75 OR CMF < -0.05

Buy & Hold (benchmark): buy at the first open and hold to the end of the window.

2. Performance comparison

Strategy	Total Return	CAGR	Volatility	Sharpe	Sortino	Max Drawdown	Win Rate	# Trades
Buy & Hold	344.3%	34.6%	27.4%	1.04	1.79	-23.1%	n/a	0
Trend Following	31.3%	5.6%	10.7%	0.18	0.29	-17.7%	47%	19
Mean Reversion	68.9%	11.0%	8.8%	0.75	1.50	-7.2%	100%	5
Custom Triple Confirmation	11.9%	2.3%	10.0%	-0.13	-0.19	-18.9%	55%	22

Highest Sharpe ratio: **Buy & Hold** (1.04).

3. Equity curves

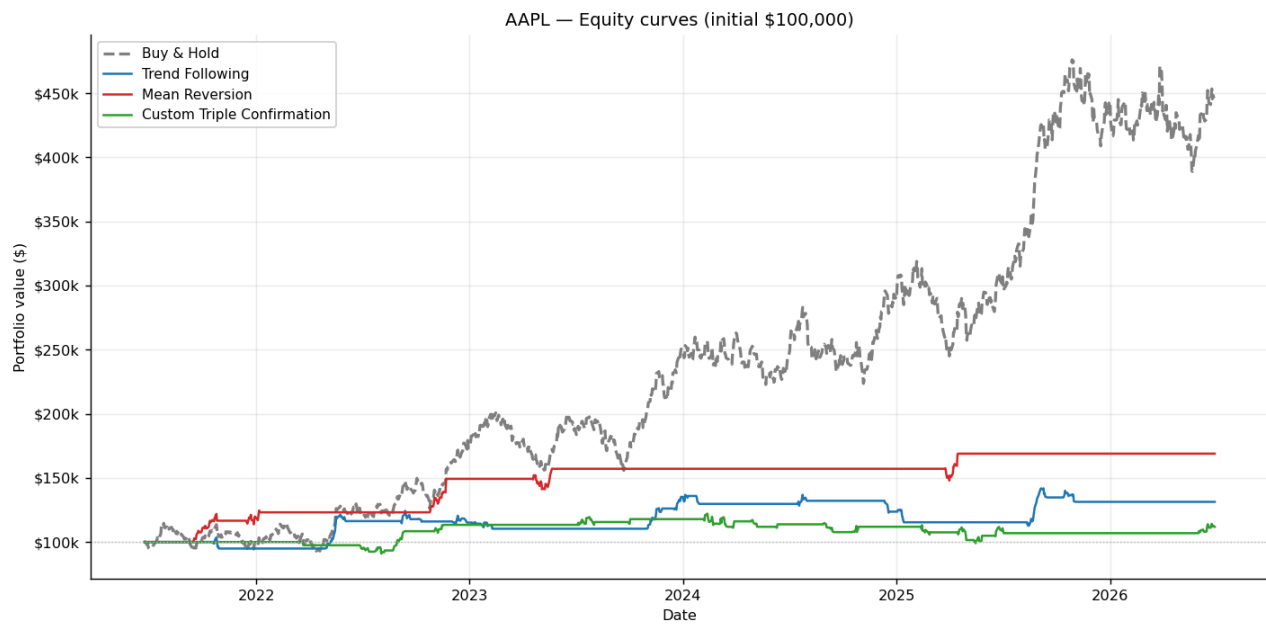


Figure 1. Portfolio value over time vs Buy & Hold.

4. Drawdowns

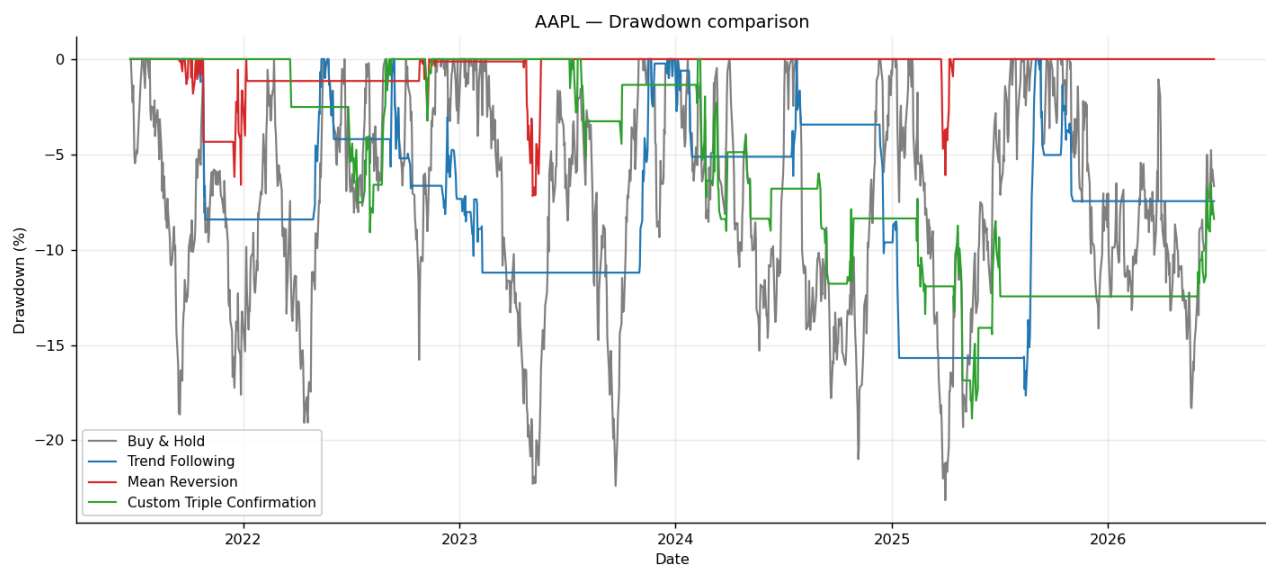


Figure 2. Underwater plot — depth and duration of losses from prior peaks.

5. Price charts with signals



Figure 3. Trend Following: price, SMA-50/200, Bollinger Bands and trade markers (RSI below).



Figure 4. Mean Reversion: price, SMA-50/200, Bollinger Bands and trade markers (RSI below).

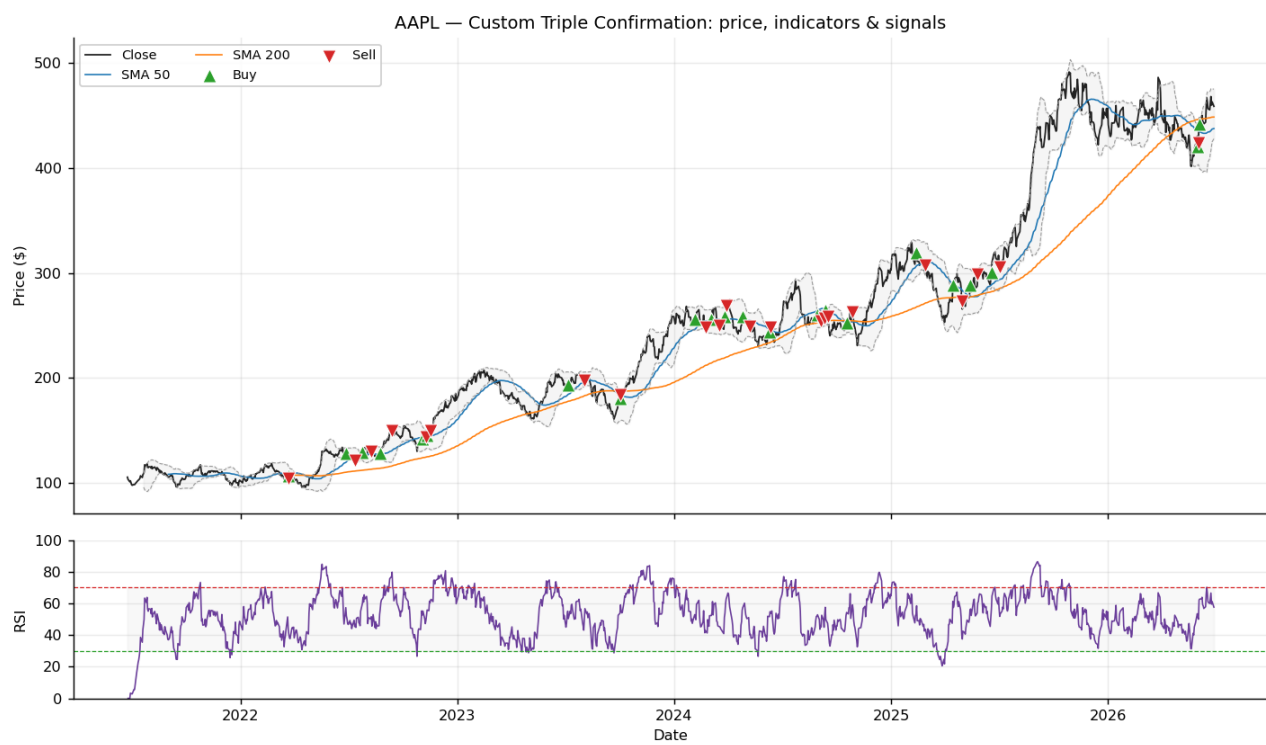


Figure 5. Custom Triple Confirmation: price, SMA-50/200, Bollinger Bands and trade markers (RSI below).

6. Discussion of results

Over the 3 active strategies tested on AAPL, **Buy & Hold** delivered the strongest risk-adjusted performance with a Sharpe ratio of 1.04, which is the headline answer to the project question: on a risk-adjusted basis this strategy was the best performer in this sample.

On raw total return, **Buy & Hold** led at 344.3%, versus the Buy & Hold benchmark at 344.3%. Higher absolute return does not always coincide with the best Sharpe: an aggressive strategy can win on return while taking on volatility that a risk-adjusted lens penalises.

For capital preservation, **Mean Reversion** had the shallowest maximum drawdown (-7.2%). The active strategies spend meaningful time in cash, so they typically cut drawdowns relative to always-invested Buy & Hold — the trade-off is that they also miss part of sustained up-moves, which is visible in the equity curves.

Reading the strategies individually: the trend-following rules only engage when MACD, ADX and the moving-average filter agree, so it sits out choppy ranges and concentrates risk in established trends. Mean reversion does the opposite — it buys statistically stretched selloffs (RSI<30 and a close below the lower Bollinger Band), which fires rarely and can underperform badly when a downtrend keeps going rather than reverting. The custom triple-confirmation strategy is the most selective: it only buys pullbacks inside a confirmed uptrend (EMA structure), with momentum turning up (RSI crossing 50) and volume confirming accumulation (CMF>0).

Caveats: results are sensitive to the chosen window and parameters (thresholds, lookbacks), a single ticker is a small sample, and these backtests exclude taxes and assume fills at the next open with only a 1 bp slippage allowance. A robust conclusion would repeat this across many tickers and time periods and test parameter stability.

Note: this run used the synthetic data fallback because Alpaca credentials/network were unavailable in this environment. Numbers are illustrative of the pipeline; set ALPACA_API_KEY and ALPACA_SECRET_KEY and re-run for live market data.

Disclaimer: This is an educational backtesting exercise, not investment advice. Backtested results are hypothetical, ignore taxes and some market frictions, and do not guarantee future performance.